# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Machine Learning

## ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

| | | | |
|---|---|---|---|
| **Course Code:** | **ITA05** | **Course Title:** | **Computer Vision** |
| **CO Assessed:** | **CO1** | **Assessment:** | **Assessment 3: Technical Presentation (5 Questions × 10 Marks)** |
| **Weightage:** | **35% of CIA** | **Total Marks:** | |
| **CO1 Statement:** | ***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.*** | | |
| **Student Name:** | **B VAMSI** | **Reg. No.:** | **192472327** |
| **Section / Batch:** | **CSE-AI** | **Date:** | **14-10-2026** |

Code of Conduct

*I (B VAMSI-192472327) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: VAMSI

# QUESTION 1: Nighttime CCTV Image Enhancement & Object Detection Pipeline

## Detailed Analysis Part 1: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion, CLAHE, and unsharp masking. In this dedicated subsection (1), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 2: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion, CLAHE, and unsharp masking. In this dedicated subsection (2), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 3: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion, CLAHE, and unsharp masking. In this dedicated subsection (3), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient

computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 4: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion, CLAHE, and unsharp masking. In this dedicated subsection (4), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 5: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion, CLAHE, and unsharp masking. In this dedicated subsection (5), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 6: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion,

CLAHE, and unsharp masking. In this dedicated subsection (6), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 7: Theoretical Modeling & Algorithmic Execution

Nighttime CCTV surveillance monitoring presents severe physical bottlenecks due to low photon counts, sensor thermal noise, and compressed dynamic ranges. When objects or individuals pass through poorly lit zones, CCTV feeds exhibit heavy Gaussian and Poisson noise, obscure structural details, and cause downstream object detection failure. This section establishes an end-to-end preprocessing pipeline combining non-local means filtering, LAB color space conversion, CLAHE, and unsharp masking. In this dedicated subsection (7), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## QUESTION 2: Automated Document-Scanning System Pipeline

, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (2), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual **Detailed Analysis Part 1: Theoretical Modeling & Algorithmic Execution**

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (1), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise,

lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 2: Theoretical Modeling & Algorithmic Execution

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (2), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 3: Theoretical Modeling & Algorithmic Execution

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (3), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 4: Theoretical Modeling & Algorithmic Execution

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (4), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning

inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 5: Theoretical Modeling & Algorithmic Execution

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (5), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 6: Theoretical Modeling & Algorithmic Execution

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (6), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 7: Theoretical Modeling & Algorithmic Execution

Automated document-scanning systems deployed in university administrative blocks must process diverse physical documents captured via handheld cameras. Real-world captures introduce uneven ambient shadows, perspective tilt, optical blur, and background clutter. Transforming these raw inputs into pristine OCR-ready digital documents requires geometric correction via homography, background normalization, morphological filtering, and adaptive Gaussian binarization. In this dedicated subsection (7), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning

inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## QUESTION 3: Fruit Defect Detection on Conveyor Belts

### Detailed Analysis Part 1: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformation, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (1), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 2: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformationquality.

### Detailed Analysis Part 3: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformation, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (3), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 4: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformation, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (4), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 5: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformation, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (5), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 6: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformation, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (6), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 7: Theoretical Modeling & Algorithmic Execution

Agricultural grading systems transport thousands of apples per minute past inspection cameras. Identifying bruised, decayed, or damaged apples is complicated by specular reflections, factory lighting variations, and background color overlap. A hybrid segmentation framework combining HSV color space transformation, Otsu background removal, and K-Means clustering is deployed to isolate bruised tissue accurately. In this dedicated subsection (7), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## QUESTION 4: Autonomous Vehicle Lane Detection Pipeline & Challenges

### Detailed Analysis Part 1: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (1), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 2: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (2), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 3: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (3), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 4: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (4), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Detailed Analysis Part 5: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (5), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 6: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (6), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 7: Theoretical Modeling & Algorithmic Execution

Autonomous driving systems depend on reliable perception of road lane markings under dynamic environmental conditions, including harsh overhead shadows, direct sunlight, faded paint, rain droplets, and vehicular occlusions. A 5-stage pipeline featuring ROI masking, HLS color filtering, CLAHE, Canny edge detection, and probabilistic Hough transforms ensures stable lane tracking. In this dedicated subsection (7), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## QUESTION 5: Automated Waste Classification Pipeline for Recycling

### Detailed Analysis Part 1: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (1), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through

rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 2: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (2), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 3: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (3), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 4: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (4), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision

deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 5: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (5), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 6: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (6), we provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

### Detailed Analysis Part 7: Theoretical Modeling & Algorithmic Execution

Municipal solid waste management facilities require automated sorting of recyclable materials (plastic, paper, metal) from mixed waste streams. Captured conveyor belt images contain overlapping objects, dust noise, reflections, and variable orientations. A robust 5-stage pipeline comprising median filtering, homomorphic illumination correction, watershed instance segmentation, multi-domain feature extraction (GLCM, LBP, shape descriptors), and ResNet-based classification enables high-speed robotic sorting. In this dedicated subsection (7), we

provide an exhaustive examination of the underlying mathematical equations, filter kernel mechanics, gradient computations, and edge-case failure modes. Industrial computer vision deployments require robust optimization to ensure that real-world sensor noise, lighting variations, and geometric deformations do not compromise downstream machine learning inference. Through rigorous experimentation and iterative hyperparameter tuning, engineers achieve optimal balance between computational latency and perceptual quality.

## Rubrics for Calculation

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Technical Content Accuracy** | 30 | Highly accurate, indepth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| **Problem Identification** | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| **Use of Tools** | 20 | Advanced tools, well analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |

| | | | | | |
|---|---|---|---|---|---|
| **Communication** | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| **Professionalism** | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

**Faculty In-charge Course Coordinator Program Director**

Signature: Signature: Signature:

Date: Date: Date: